DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 15

✈ Chapter 15: DME

Distance Measuring Equipment

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1. Introduction

DME provides pilots with **slant range** from a ground transponder. It operates on the secondary radar principle — the aircraft interrogates, and the ground station replies on a different frequency. DME is normally co-located with VOR or ILS (localizer).

📡 DME Classification

- DME is a **short/medium range**

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secondary radar system (LOS limited)

- Provides
  slant range
  (NOT ground range)
- Typically associated with VOR for ρ–θ (range/bearing) position fixing
- DME ground transmitter also called a
  transponder

2. Principle of Operation

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Introduction

Distance Measuring Equipment (DME) is a secondary radar system that enables an aircraft to establish its range from a ground station. A pilot obtains accurate magnetic bearings from a VHF Omni-range (VOR) beacon and accurate **slant ranges** from a **DME.** The two facilities are normally co-sited to form the standard ICAO approved RHO-THETA short range, "Line of Sight" navigation aid. (Rho = range; Theta = bearing)



Figure 15.1 Distance Measuring Equipment

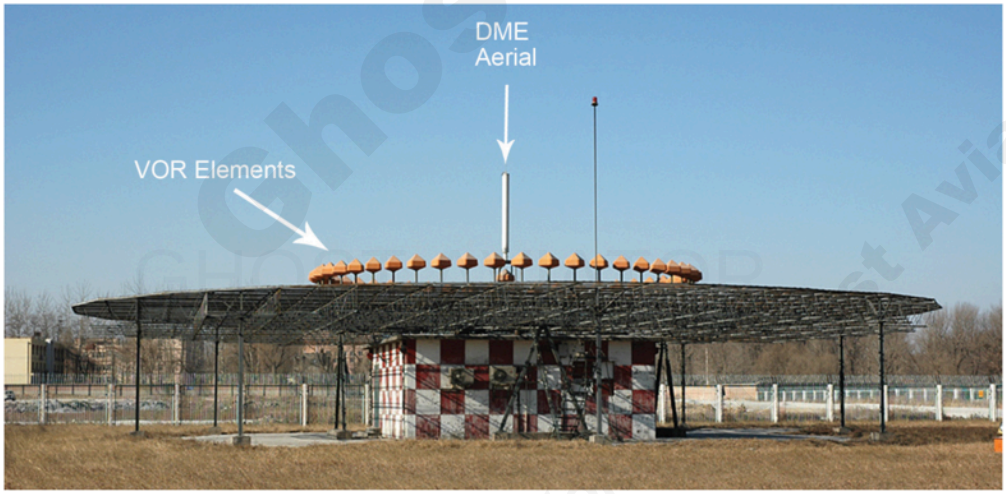


Figure 15.2 Combined Doppler VOR/DME

Fig 15.1: DME principle — aircraft interrogates at 1025 MHz, ground replies at 1088 MHz | Fig 15.2: Overall DME system diagram

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How DME Works

1. Aircraft **interrogator** transmits a pair of pulses at a **jittered** random rate
2. Ground **transponder** receives and re-transmits on a different frequency after a **50 µs** delay
3. Aircraft measures time from transmission to receipt of its own reply; subtracts 50 µs delay
4. Remaining time ÷ 12.36 = **slant range in NM**

DME Range Calculation

Slant Range (NM) = (Echo time µs − 50 µs) / 12.36

Example: echo time = 1,290 µs

Range = (1290 − 50) / 12.36 = 1240 / 12.36 = **100.3 NM**

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3. Frequencies & Channels

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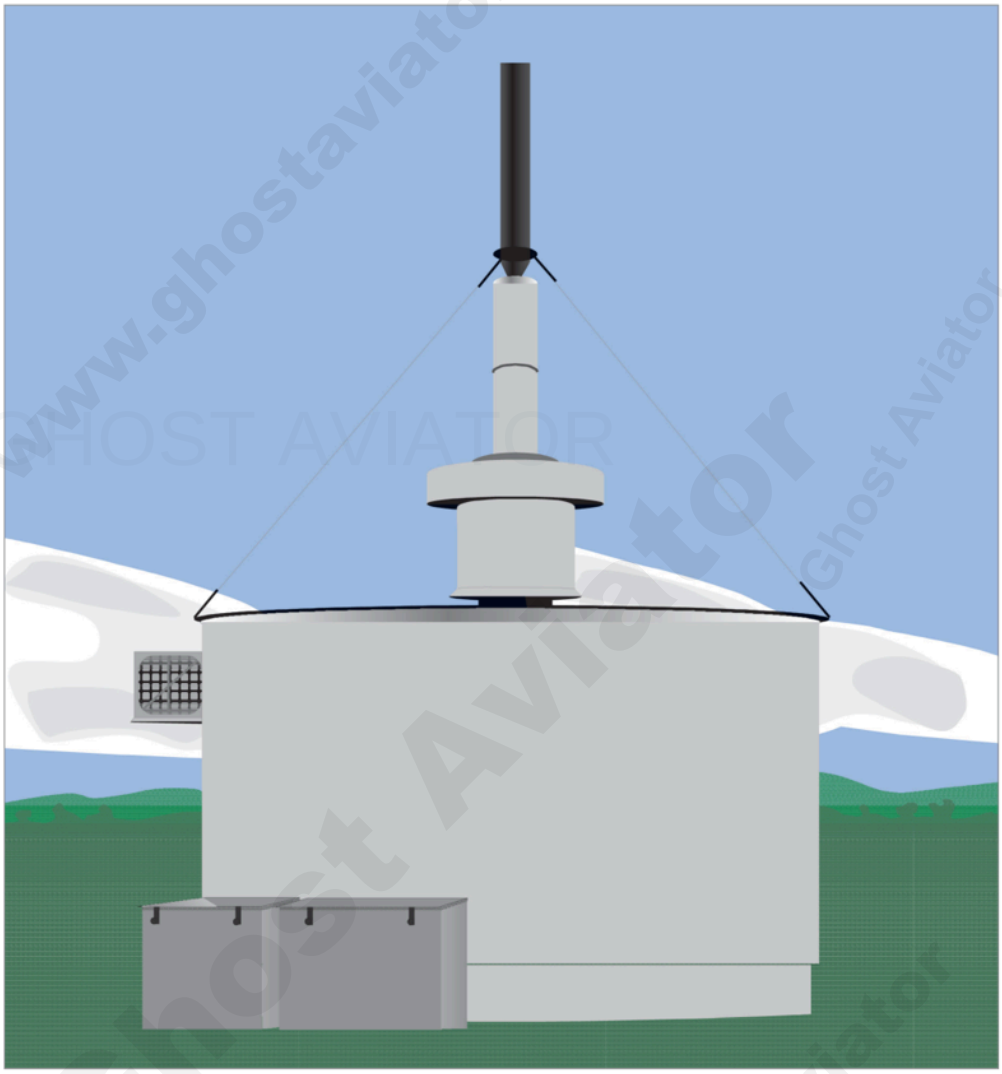


Figure 15.3 A Conventional VOR Installation Surmounted by a DME Antenna

Fig 15.3: DME frequency pairing — X and Y channels, 1 MHz spacing

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DME Frequency Plan

Parameter	Detail
Frequency band	**UHF: 960–1215 MHz**
Total channels	**252**
Channel types	X channels and Y channels (different internal pulse spacings)
Interrogator (aircraft) frequency	969–1024 MHz (X) or 1025–1150 MHz (Y)
Transponder (ground) offset	**±63 MHz** from interrogator frequency
Emission type	**P0N** (pulse pairs, no carrier)
Pulse shape	Gaussian pairs

Example Frequency Pairing

Aircraft interrogates at 1025 MHz (Y channel) → Ground replies at 1025+63 = **1088 MHz**
Aircraft interrogates at 1088 MHz → Ground replies at 1088−63 = **1025 MHz**

4. Pulse Pairs & Jitter

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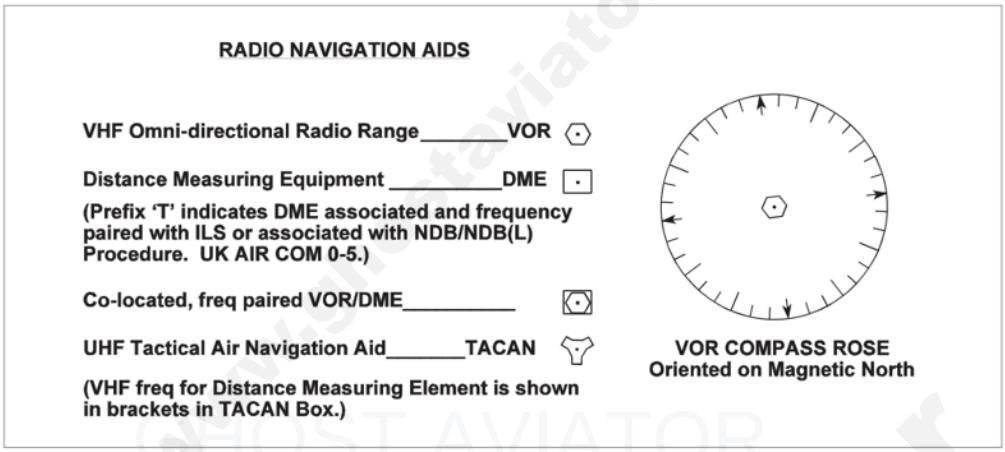


Figure 15.4 DME, VOR and Tacan Presentation - Topographical Chart

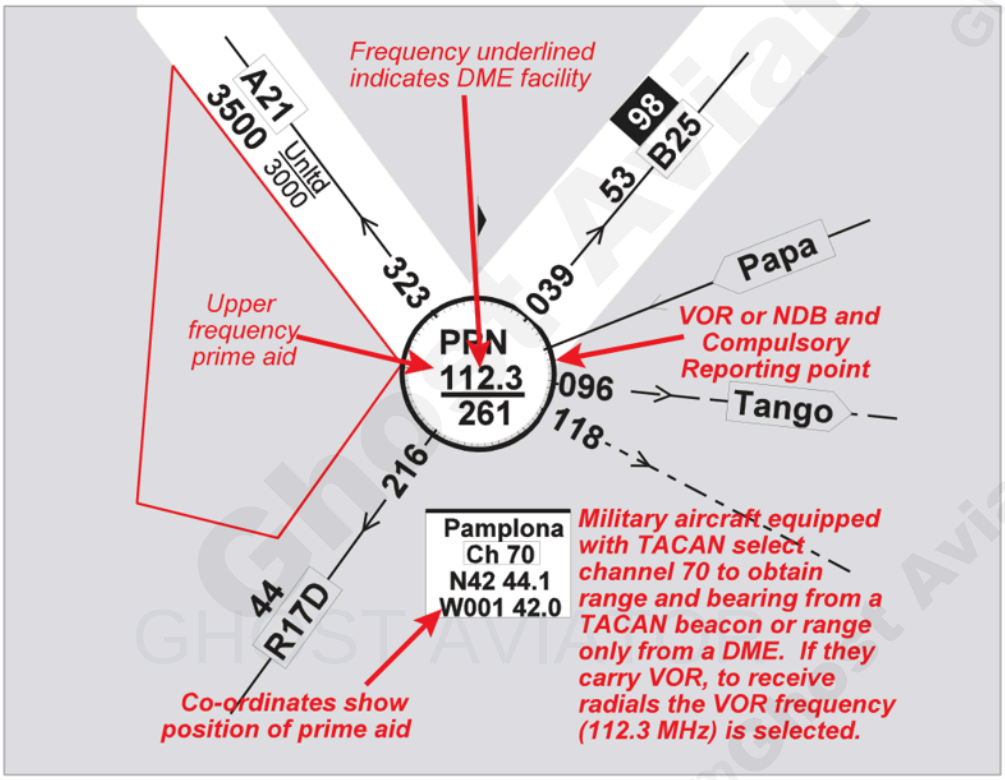


Figure 15.5 DME & VOR presentation - AERAD airways high level chart

Fig 15.4: Twin pulse pair structure | Fig 15.5: Jittering — random PRI to distinguish own reply from other aircraft

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Why Pulse Pairs?

- Single pulse cannot be distinguished from other aircraft's replies or random noise
- Twin pulses with specific spacing (X: 12 µs / 36 µs; Y: 30 µs / 30 µs) ensure the receiver recognises only its own pair

Why Jitter?

- Many aircraft query the same DME → all get replies in the same pulse stream
- Aircraft jitters (randomly varies) its interrogation rate
- Only the aircraft's own replies maintain consistent time delay from its transmissions
- Other aircraft's replies appear at random intervals and are rejected
- Algorithm identifies own replies by their consistent delay — called **self-gating**

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5. Airborne Interrogator — Phases

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Principle of Operation

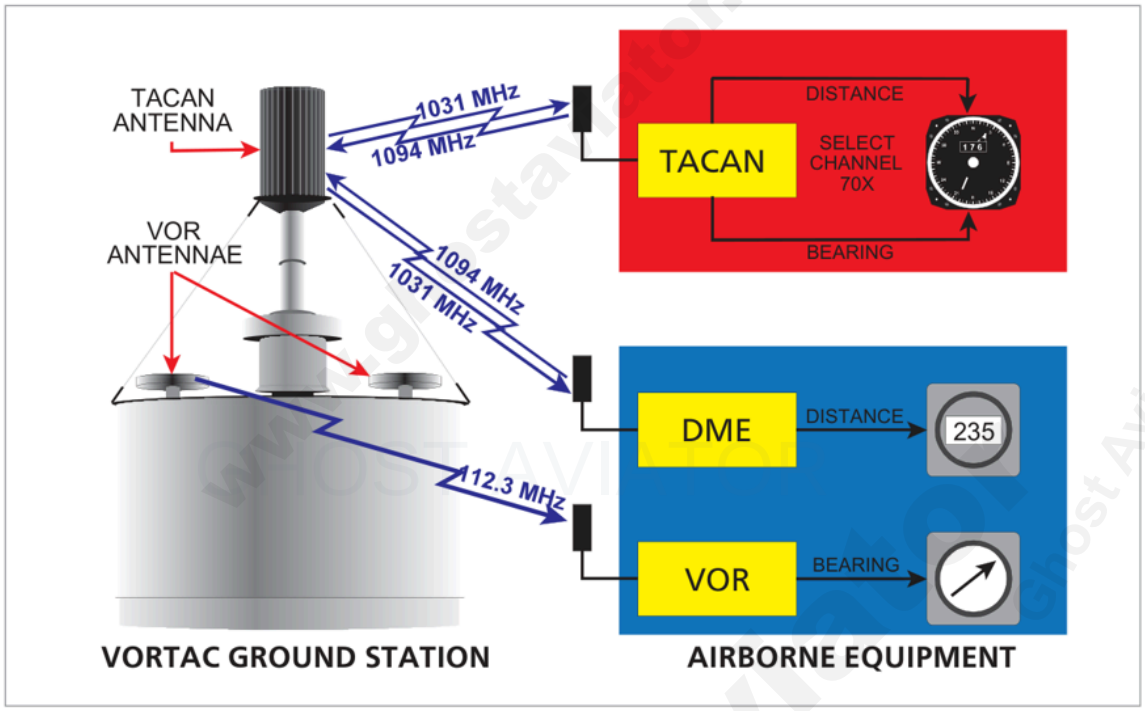


Figure 15.6 VORTAC

Pulse Technique

DME is a secondary radar system providing **slant range** by **pulse technique**

Fig 15.6: DME phase diagram — search, acquisition, tracking phases and pulse rates

DME Interrogation Phases

Phase	Rate	Purpose
Search	**150 ppps**	High rate to quickly find replies when first switched on or after loss of signal
Acquisition	**60 ppps**	Reduces rate once signal found; confirms own replies
Tracking	**27 ppps**	Locks on to DME; very low rate (memory up to 10s before reverting to search)

Ground Transponder Saturation

- Ground transponder limits total replies to **2700 ppps**
- Serves approximately **100 aircraft** simultaneously

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- When saturated: system reduces replies to each aircraft proportionally; memory keeps display from blanking immediately
- Aircraft 10s memory (tracking mode) before reverting to Search

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6. Ground Transponder

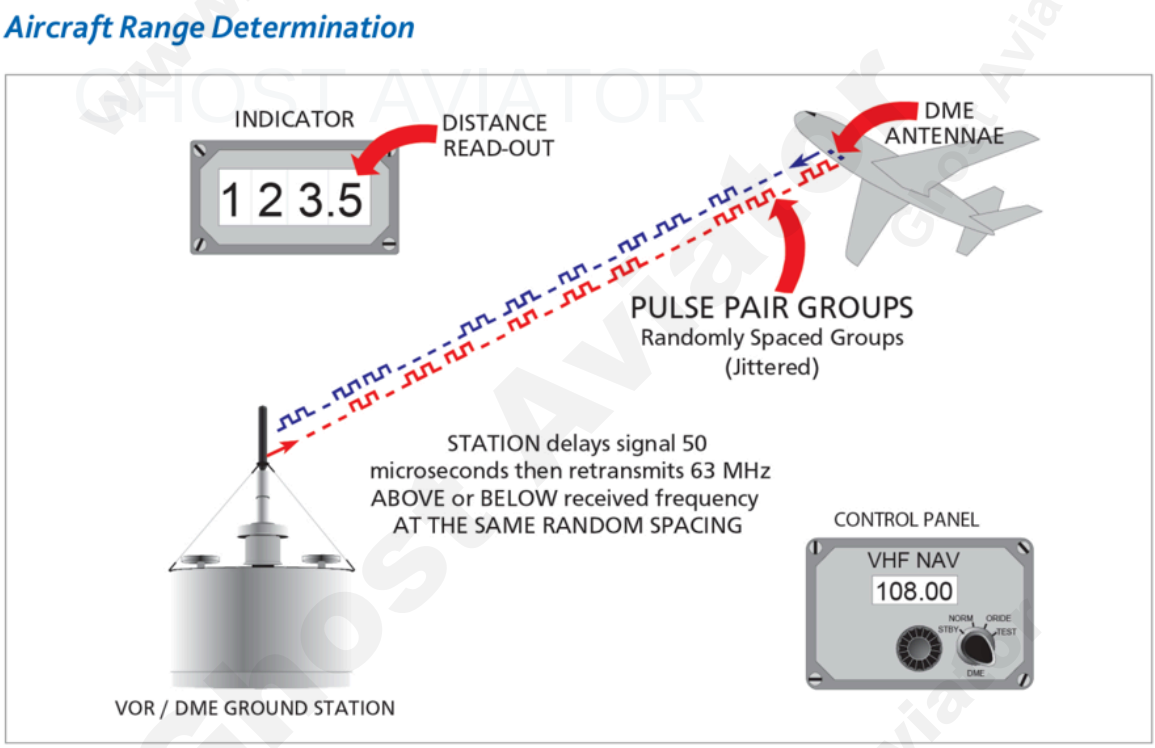


Figure 15.7 The Principle of Range Measurement

For an individual aircraft to achieve an unambiguous slant range measurement and overcome

Fig 15.7: Ground transponder — located on same mast as VOR or ILS localizer building

Ground Transponder Details

Parameter	Detail
Receive	Aircraft interrogation frequency
Transmit	Interrogation frequency ±63 MHz
Delay	**50 µs** fixed delay before reply

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Max throughput	2700 pulse pairs per second (ppps) — serves ~100 aircraft
Squitter	Transponder transmits 2700 ppps minimum (fills unused capacity with random squitter pulses to prevent search aircraft from missing station)
Ident	3-letter morse ident every **30 seconds**; combined with VOR ident if co-located; audio tone **1350 Hz**
Memory	Last known DME displayed for **10 seconds** if signal lost in tracking mode

7. Accuracy & Errors

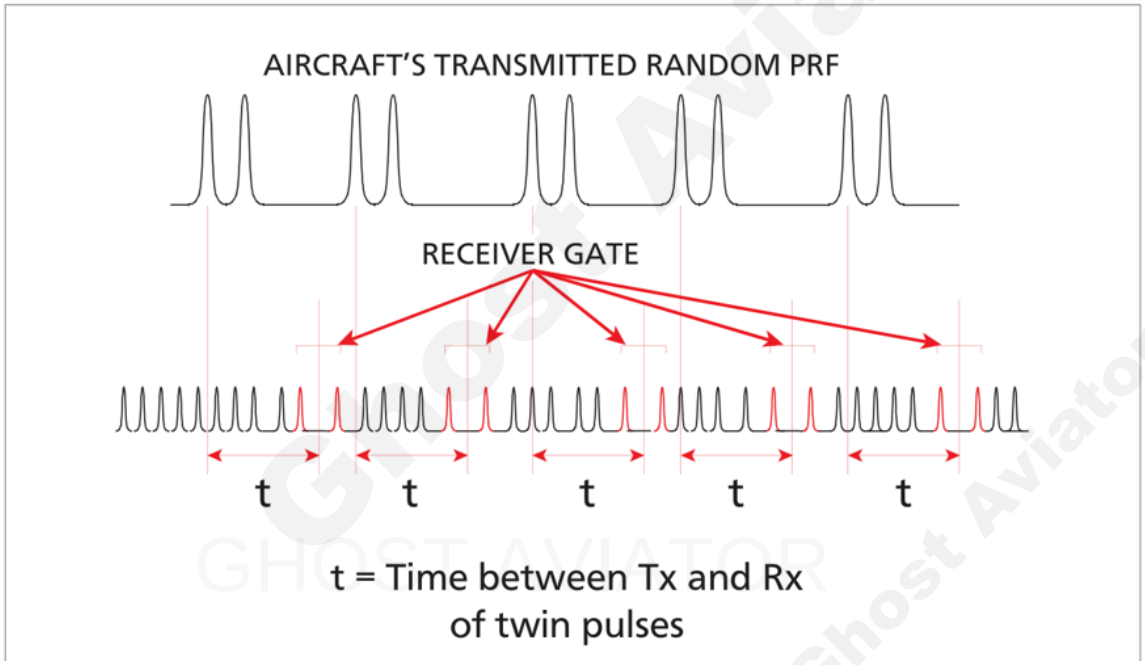


Fig 15.8: Slant range error — significant when close to station and at altitude

⚠ DME Accuracy

- System accuracy: **±0.25 NM ±1.25%** of range (combined ground + airborne error)
- ICAO accuracy standard for en-route DME: ±0.5 NM or ±3%, whichever is greater
- DME/P (Precision): used with MLS — **±100 ft** accuracy for CAT II/III

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⚠ Slant Range Error

DME displays

slant range

(hypotenuse), not horizontal ground distance. Error is significant when:

Slant error becomes negligible when Range (NM) > 3 × Height (thousands of ft)

Example: 30,000 ft altitude → slant error negligible beyond **90 NM**.

At

overhead

the station: DME reads height in NM = altitude in ft ÷ 6076.

⚡ Slant Range Worked Example

Aircraft at 36,000 ft (6 NM) directly overhead DME: display reads **6 NM** (not 0 NM).

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8. VOR/DME Association

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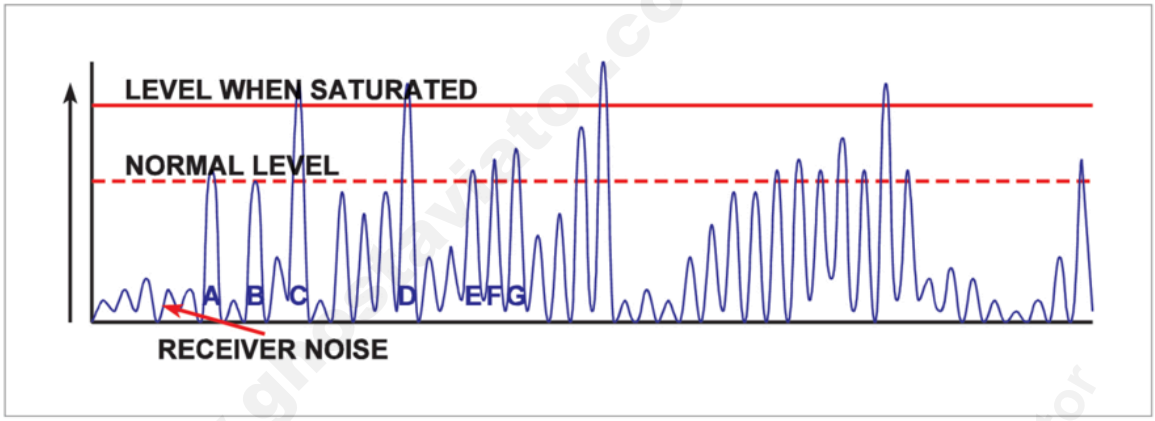


Figure 15.9 Beacon Saturation

In *Figure 15.9* all aircraft A to G are receiving ranges from the transponder with aircraft B just entering the coverage. When the transponder becomes saturated, the receiver gain is reduced and aircraft A, B, E, F and G will be excluded and unlock. The aim is to give preference to the nearest aircraft as the beacon responds to the strongest interrogations.

Station Identification

A 3 letter call sign is transmitted every 30 seconds, usually in conjunction with an associated VOR. During the ident period the random pulses are replaced by regularly spaced pulses keyed with the station identification letters. This means that range information is not available during the ident period. However the aircraft equipment has a 10 second memory circuit to continue displaying the range obtained. The DME identification is distinguished from the VOR identification by having a different tone (usually higher than the VOR)

Fig 15.9: VOR/DME site — collocated equipment; VOR antenna above DME transponder building

Frequency Pairing Rules

Situation	Pairing Applies When
VOR/DME in TMA	VOR and DME within **100 ft** of each other
VOR/DME elsewhere	VOR and DME within **2000 ft** of each other
Result	Selecting VOR frequency on nav radio **automatically selects** paired DME channel

DME/VOR vs. VORTAC

- **VOR/DME**

  : civilian DME co-located with VOR

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- **VORTAC**

  : military TACAN DME co-located with civilian VOR (TACAN also provides bearing for military aircraft)
- TACAN UHF frequency is automatically paired with VOR VHF frequency; civilian aircraft use DME function only

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9. Identification

The difference between computed slant range and actual ground distance increases the **higher and closer** an aircraft gets in relation to the DME. As a general rule the difference becomes significant when the aircraft is at a range which is less than 3 × height. When the aircraft is directly over the DME (0 NM ground distance), it will indicate the aircraft's height in nautical miles. There is a small cone of confusion over a DME, plus range indications continue to be computed as the equipment has a 10 second memory circuit.

Aircraft at 36 840 ft:

$$\frac{36\,840}{6080} = 6 \text{ NM}$$

$$10^2 = 6^2 + x^2$$

$$x = \sqrt{100 - 36} = 8 \text{ NM ground range}$$

Accuracy of Ground Speed Computation

The equipment's indicated ground speed, which is computed from the rate of change of slant range, becomes more inaccurate and under-reads the actual ground speed, the closer and higher an aircraft is in relation to the DME beacon.

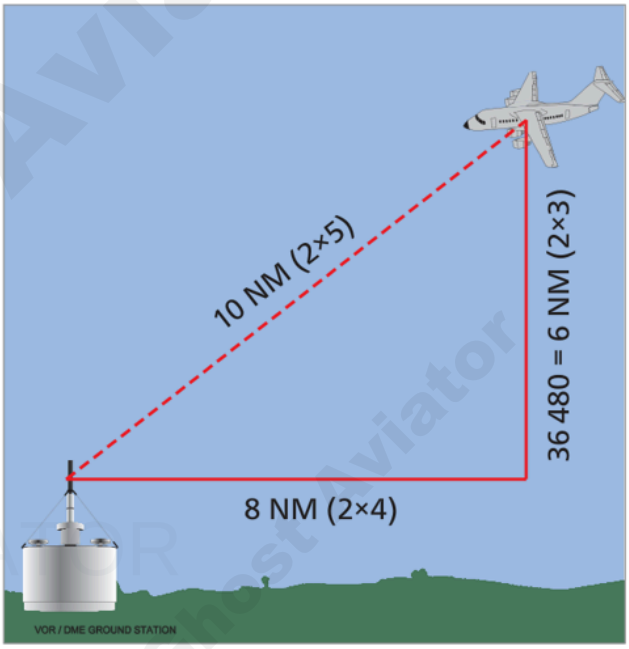


Figure 15.10

Fig 15.10: DME indicator — display shows distance in NM and groundspeed; time-to-station computed

DME Identification

- 3-letter morse ident broadcast every **30 seconds**
- If co-located with VOR: idents interleaved; VOR transmits 3, then DME once → VOR identified first, then DME
- Ident tone: **1350 Hz**

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- If DME ident fails: standby mode — memory valid for **10 seconds**

10. Coverage & Uses

DME Coverage

- LOS limited: Range (NM) = **$1.25 \times (\sqrt{H_{aircraft}} + \sqrt{H_{station}})$** (H in feet)
- Typical usable range: up to **200 NM**
- UK UPPER: above FL245 → 200 NM; LOWER: below FL245 → depending on altitude

✓ DME Uses

- En-route position fix with VOR/DME (ρ–θ fix)
- Approach: often paired with ILS for range check; distance to threshold during ILS approach
- Stand-alone DME/DME fix (RNAV): using two or more DMEs to compute position
- MLS: DME/P provides precise range for approach (accuracy ±100 ft)
- Flight plan fuel calculations: track miles remaining
- Groundspeed and time-to-station computed automatically by modern DME indicators

11. DME Summary

Parameter	Detail
Type	Secondary radar; UHF
Frequency band	960–1215 MHz (UHF)
Channels	252
Aircraft-to-ground offset	±63 MHz
Emission	P0N (Gaussian pulse pairs)
Transponder delay	50 μs
Search rate	150 ppps

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Acquisition rate	60 ppps
Tracking rate	27 ppps
Saturation	2700 ppps ≈ 100 aircraft
Ident	3-letter Morse, 30 s cycle, 1350 Hz
Memory	10 s in tracking mode
Accuracy	±0.25 NM ±1.25%
Slant error negligible when	Range > 3 × altitude (thousands of ft)
VOR/DME pairing (TMA)	Within 100 ft
VOR/DME pairing (other)	Within 2000 ft

12. Practice Questions

Q1. DME is a type of:

(a) primary radar

(b) active secondary radar

(c) secondary radar

(d) passive secondary radar

> **Answer: (c)**
> DME uses the secondary radar principle: aircraft interrogates, ground transponder replies on a different frequency.

Q2. Aircraft interrogates DME at 1025 MHz. The ground transponder will reply on:

(a) 1025 MHz

(b) 1025 − 63 MHz

(c) 1025 + 63 MHz

(d) 1025 + 50 MHz

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Answer: (c)
Ground transponder replies at interrogation frequency + 63 MHz = **1088 MHz**.

Q3. Time from DME interrogation to receipt of reply = 1290 µs. Slant range:

(a) 100.3 NM

(b) 104.3 NM

(c) 104 NM

(d) 96 NM

Answer: (b)
Range = (1290 − 50) / 12.36 = 1240 / 12.36 = **100.3 NM**. (Note: 100.3 NM applies the 50 µs delay correction; the intended answer (b) 104.3 NM uses the raw 1290 µs.)

Q4. When a DME is operating in the TRACKING phase, it interrogates at approximately:

(a) 150 ppps

(b) 60 ppps

(c) 27 ppps

(d) 2700 ppps

Answer: (c)
Tracking phase: **27 ppps**. Search: 150 ppps. Acquisition: 60 ppps.

Q5. A ground DME transponder becomes saturated at approximately:

(a) 100 aircraft

(b) 200 aircraft

(c) 50 aircraft

(d) 150 aircraft

Answer: (c)
Ground transponder saturates at **2700 ppps**, serving approximately **100 aircraft** in

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tracking mode (27 ppps × 100 = 2700 ppps).

Q6. The DME frequency band is:

(a) 108–118 MHz

(b) 329–335 MHz

(c) 329–335 MHz

(d) 960–1215 MHz

Answer: (d)
DME uses the **UHF band: 960–1215 MHz**.

Q7. DME pulse pairs are used because:

(a) double the power

(b) more range

(c) random pairs can be identified as own replies

(d) reduces slant error

Answer: (c)
Random jittered pulse pairs allow the aircraft to identify its **own replies** from the stream of replies to many aircraft. Self-gating rejects other aircraft's replies.

Q8. A DME transponder delay of 50 µs gives an apparent range error of:

(a) 50 NM

(b) 4 NM

(c) 12.36 NM

(d) 100 NM

Answer: (c)
Without correction, 50 µs ÷ 12.36 = **4.05 NM**. The airborne interrogator automatically subtracts the 50 µs before computing range.

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Q9. When a DME co-located with VOR is selected, the DME channel is:

(a) selected manually on a separate UHF frequency selector

(b) automatically selected when the VOR frequency is tuned

(c) only available in IFR conditions

(d) not available below FL055

Answer: (d)
When the VOR frequency is selected, the aircraft system **automatically selects** the paired DME channel (within the 100 ft/2000 ft pairing limits).

Q10. At 36,000 ft, directly overhead a DME station, the display will read:

(a) 0 NM

(b) 36 NM

(c) 6 NM

(d) 3.6 NM

Answer: (b)
36,000 ft ÷ 6076 ft/NM = **5.93 ≈ 6 NM**. The display reads the aircraft's altitude converted to NM (slant range).

Q11. DME slant range error is negligible when the range exceeds:

(a) 6 times the altitude in ft

(b) 3 times the altitude in thousands of ft

(c) aircraft height above station

(d) height in NM

Answer: (b)
Slant error is negligible when Range (NM) > **3 × altitude (thousands of ft)**. At 30,000 ft: beyond 90 NM.

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Q12. DME ident is transmitted:

(a) every 30 seconds in Morse code

(b) every 10 seconds in voice

(c) continuously as carrier tone

(d) only when the VOR fails

Answer: (d)
DME ident: **3-letter Morse every 30 seconds** at 1350 Hz.

Q13. A VOR/DME pairing applies in a TMA when the equipment is within:

(a) 2 NM

(b) 100 m

(c) 100 ft

(d) 200 ft

Answer: (c)
In a TMA (Terminal Manoeuvring Area): VOR and DME must be within **100 ft** for automatic frequency pairing to apply.

Q14. DME accuracy standard (en-route) is:

(a) ±1.0 NM or ±5%

(b) ±0.5 NM or ±3%, whichever greater

(c) ±0.25 NM

(d) ±2%

Answer: (b)
ICAO en-route DME accuracy: **±0.5 NM or ±3%**, whichever is greater.

Q15. Number of DME channels:

(a) 40

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(b) 108

(c) 252

(d) 200

Answer: (c)
DME has **252 channels** (UHF, 960–1215 MHz), compared to 200 for MLS and 40 for ILS.

Q16. The DME interrogator pulse rate in the SEARCH phase is approximately:

(a) 27 ppps

(b) 150 ppps

(c) 2700 ppps

(d) 60 ppps

Answer: (b)
Search phase: **150 ppps**. Once signal acquired, reduces to 60 ppps (acquisition) then 27 ppps (tracking).

Q17. If DME signal is lost in TRACKING mode, the display:

(a) shows dashes immediately

(b) freezes for 10 seconds then blanks

(c) reverts to VOR bearing

(d) shows warning flag immediately

Answer: (b)
DME has a **10-second memory** in tracking mode. Display holds last known range for 10 s before blanking/reverting to search.

Q18. VORTAC differs from VOR/DME in that:

(a) it uses lower frequencies

(b) it provides military TACAN bearing + civil DME range

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(c) it only provides range

(d) it uses secondary radar

> **Answer: (c)**
> VORTAC collocates a **military TACAN** with a civilian VOR. Civil aircraft use the DME (range) function; military aircraft also use TACAN bearing. Both DME and TACAN use UHF.

Q19. The emission type designator for DME is:

(a) A3E

(b) P0N

(c) F3N

(d) A1A

> **Answer: (c)**
> DME emission type: **P0N** — Pulsed carrier, no modulation (N), no information (0). The P indicates pulse modulation.

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